

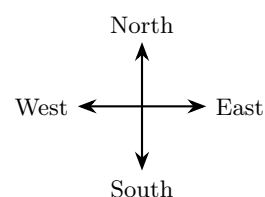
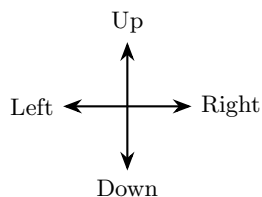
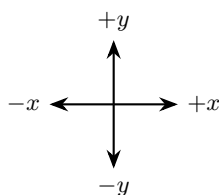
Instructions: Answer all questions. Show your work for each problem by following the four problem-solving steps below.

Problem-Solving Steps

1. What am I given? What am I finding?
2. What equation connects them?
3. Plug in and solve.
4. Check the units and direction (if applicable).

Kinematics Formulas

$$\bar{v} = \frac{\Delta x}{t} \quad s = \frac{d}{t} \quad \Delta x = x - x_0 \quad a = \frac{\Delta v}{t} = \frac{v - v_0}{t}$$

Vector Directions**Hints**

If a vector has no given direction, use positive (+) or negative (-).

starts from rest $\rightarrow v_0 = 0$. *comes to rest* $\rightarrow v = 0$.

1. A cyclist rides 60 km in 3 h.
(a) What is her speed in $\frac{\text{km}}{\text{h}}$?

- (b) What is her speed in $\frac{\text{m}}{\text{s}}$?

2. A runner moves 800 m north in 40 s. What is her average velocity?

3. A runner jogs 400 m in 100 s, then walks 200 m in 50 s. What is her average speed for the whole trip?

